ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** spaces between particles	Particles of one fill the gaps.
3	**(b)** temperature increases the motion of particles	Faster particles.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** change of state	Latent heat.
6	**(d)** equal at equilibrium	Dynamic equilibrium.
7	**(c)** A is true but R is false.	A is true; R is false: heat is used to overcome forces between particles.
8	**(d)** Particles have the same kinetic energy but different arrangement	Same temperature, different state.
9	**(d)** oxygen	Gas.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gas particles move rapidly in all directions (1); They spread uniformly and fill the container (1).	1+1
12	Evaporation is slow (1); Little cooling effect (1).	1+1
13	Only particles at the surface with enough energy escape (1); It occurs only at the liquid surface (1).	1+1
14	Reading stays at 0 degrees Celsius (1); Heat is used to change the state, not to raise temperature (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	-10 to 0: ice warms (1); 0 to 100: melting at 0 then water warms (1); At 100: boiling then steam warms to 110 (1).	1+1+1
16	Ice: least energy, close and fixed (1); Water: medium (1); Steam: highest, far apart (1).	1+1+1
17	It rises slightly (1); Dissolved particles hinder escape of water particles (1); More heat needed (1).	1+1+1
	OR: Hot water particles move faster (1); Higher average kinetic energy (1); Cold water particles move slower (1).	1+1+1
18	Number of particles remains the same (1); Only arrangement changes (1); Mass is the quantity of matter (1).	1+1+1
	OR: Particles in cold objects still move (1); They have kinetic energy (1); Heat flows from hot to cold objects (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Independent: temperature of water (1); Dependent: time taken for ink to spread (1); Constants: volume, ink drop, beaker (1); Observation: spreads faster in warm water (1); Conclusion: higher temperature gives faster diffusion (1).	1+1+1+1+1
	OR: Increase = 1600 times (1); Particles are far apart in the gas (1); Strong attraction in liquid (1); Little space between liquid particles (1); Gas fills available space (1).	1+1+1+1+1
20	Liquid cools (1); Starts to solidify at the freezing point (1); Flat part: energy released as particles form a fixed structure (1); Temperature constant (1); Solid cools further (1).	1+1+1+1+1
	OR: Low temperature, high pressure favour solids and liquids (1); Dry ice sublimes at atmospheric pressure (1); Liquid CO_2 exists only at high pressure (1); Water changes with temperature at normal pressure (1); Boiling point depends on pressure (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** water evaporates and takes heat from the air	Evaporation.
2	**(a)** spread it open in warm, dry, windy place	Raise evaporation.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(b)** dissolved air coming out	Dissolved gases.
5	**(a)** sublimes, wax melts then remains	Different behaviours.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(d)** sublimation	Both sublime.
8	**(d)** particles close and in fixed positions	Order.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(d)** steam at 120 degrees Celsius	Highest temperature.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heat gently in a dish with an inverted funnel (1); Camphor sublimes and is collected; salt remains (1).	1+1
12	Raise: increase pressure (1); Lower: decrease pressure (1).	1+1
13	Extra heat is used for vaporisation (1); Boiling point stays constant at a given pressure (1).	1+1
14	Attractive forces are strong enough (1); Particles stay close but move around (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Vapour in air condenses on the cold glass (1); More vapour is present on humid days (1); Water does not come from inside the glass (1).	1+1+1
16	Liquid perfume evaporates (1); Vapour particles diffuse through the air (1); We smell them (1).	1+1+1
17	Large gaps in gases (1); Very small gaps in liquids (1); Particles in liquids resist being pushed closer (1).	1+1+1
	OR: Weaker forces between spirit particles (1); Particles escape easily (1); More heat is taken from the surface (1).	1+1+1
18	Different boiling points (1); Used in distillation (1); Vapour is condensed to collect (1).	1+1+1
	OR: More sublimation at higher temperature (1); Particles gain energy (1); Faster escape (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Steam carries latent heat of vaporisation (1); Gives heat on condensing (1); Pressure cooker raises pressure and boiling point (1); Hills: lower pressure lowers boiling point (1); Cooking time changes (1).	1+1+1+1+1
	OR: Smell reaches sooner at 35 degrees Celsius (1); Particles move faster when warm (1); Diffusion in air depends on particle motion (1); Also on air currents (1); Both show particles are in constant motion (1).	1+1+1+1+1
20	Sun's heat evaporates water (1); Vapour rises, condenses to cloud (1); Rain returns water (1); Sweat evaporation cools us (1); Fast particles leave carrying energy (1).	1+1+1+1+1
	OR: Solids hold shape: tools, buildings (1); Liquids flow: water supply, oil (1); Gases compress: cylinders, balloons (1); Particle forces explain use (1); Choice of states in design (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** particles of matter are stationary	Diffusion shows motion.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(d)** gas particles are far apart	Gaps.
4	**(a)** vapour particles collide more with the walls	More collisions.
5	**(c)** A is true but R is false.	A is true; R is false: the faster particles leave.
6	**(c)** liquid or compressed gas expands and absorbs heat	Expansion cooling.
7	**(b)** heat is first used for changing state	Latent heat.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** increases	Faster particles hit the walls harder and more often.
10	**(b)** mass of the particles	Lighter diffuse faster.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They gain kinetic energy (1); Temperature is a measure of average kinetic energy (1).	1+1
12	Plunger moves out (1); Heated gas expands as particles move faster (1).	1+1
13	Gas inside expands on heating (1); Particle speed increases and pressure increases (1).	1+1
14	Slow evaporation due to lower temperature (1); Humid air reduces it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Temperature rises (1); More evaporation, vapour pressure increases (1); Condensation also occurs on cool walls (1).	1+1+1
16	Air particles gain energy (1); Pressure increases (1); Balloon stretches (1).	1+1+1
17	Warm air increases temperature (1); Moving air carries away vapour (1); Evaporation is faster (1).	1+1+1
	OR: Dry ice sublimes to CO2 gas and cools the air (1); Water vapour in air condenses (1); Tiny droplets appear as fog (1).	1+1+1
18	Particles are tightly packed (1); They only vibrate (1); Little movement through the solid (1).	1+1+1
	OR: Water vapour in air touches cold surfaces (1); It loses energy and forms ice crystals (1); Vapour to solid deposition (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	P weakest, R strongest (1); Rate depends on forces (1); P spirit (1); Q water (1); R oil (1).	1+1+1+1+1
	OR: (i) Pressure rises as volume falls (1); (ii) Piston moves out because particles speed up (1); (iii) Pressure drops because collisions weaken (1); Particle model explains each (1); Collision frequency and speed (1).	1+1+1+1+1
20	Solid: fixed, vibrate (1); Melting: vibrations break pattern (1); Boiling: particles escape throughout (1); Condensation/freezing: loss of energy brings particles closer (1); Sublimation: direct escape from solid (1).	1+1+1+1+1
	OR: Dew (1); Cooling by sweat (1); Drying clothes (1); Camphor disappearing (1); Cloud formation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.